

Synthesis — Progress Report

Repo audit vs. Proposal Final V2 (BUSA 649) · 20 June 2026 · branch feature/onet-rag-step-2-retriever
· method: static read of app/lib/supabase/scripts/tests/docs + npm test (103 pass)

Executive summary

The application engineering is mature and genuinely works; the measured-result science and voice are essentially not started, and the production data plane (Supabase / pgvector / auth / ingestion / n8n) is scaffolded but dormant. All three modules run end-to-end today in mock mode and in real (Haiku) mode for the two LLM-scored modules.

- **Fit Analyzer** — fully real, deterministic, O*NET-grounded; 0–100 score with per-requirement evidence. Zero Claude cost.
- **Behavioural Simulator** — real JD-driven questions + STAR retrieval + 5-dim rubric scoring (Haiku verified). Text only.
- **Case Simulator** — most complete: pure 8-state FSM (probe / redirect / hint / exhibit-drip / never-skip-scoring), real per-turn + final scoring (Haiku verified), full adaptive chat UI.

Overall completion vs. full proposal ≈ 60% — app/platform ~75–80%, validation science ~10% (now built, see companion study), production data plane ~35%, voice 0%. Cost on track: ~\$0.015/behavioural, ~\$0.063/case, \$0/fit — far under the \$10 budget.

Completion by dimension

#	Dimension	Est.	Basis
1	Resume Fit Analyzer	85%	Real engine + UI + API; no upload UI / validation / postings grounding
2	Behavioural Simulator	85% / 0% voice	Real eval verified; voice not started
3	Case Simulator	90%	FSM + eval + scoring + UI complete & verified
4	O*NET / RAG / pgvector	40%	Taxonomy + in-memory retrieval real; pgvector stubbed; 21/1016; no postings
5	Mock vs real mode	90%	Dual-mode; real Haiku verified; embeddings real-path unused
6	API routes	80%	3 routes work; no persistence/auth
7	UI pages	85%	4 polished pages; no upload/voice/auth
8	Tests	90%	103 unit tests; no integration-vs-real-services
9	Deployment	90% mock / 30% real	Mock-mode live; real needs Supabase + auth

Fully done

- Fit scoring engine (deterministic, O*NET-grounded) — lib/matching.ts, lib/onet.ts
- Case FSM with all locked transition rules — lib/fsm/case-fsm.ts
- Case + Behavioural evaluators (heuristic mock / Haiku real, with fallback)
- Real-mode (Haiku) verification with measured cost — docs/real_mode_verification.md
- Mock/real dual-mode architecture; locked schema migrations 0001–0011; two-plane guard test; 103 tests; mock-mode Vercel deploy

Partially done

- **Parsers** — heuristic resume/JD + real PDF/DOCX extraction; but no $\geq 95\%$ robustness measurement, and extraction is regex, not LLM-driven as the proposal states.
- **O*NET / RAG** — taxonomy + in-memory retrieval real; pgvector path is a TODO stub; only 21 of 1,016 occupations; postings corpus not in retrieval.
- **API routes / UI** — work, but no persistence, no auth, no file-upload UI, no voice control.
- **Embeddings (BGE)** — full code path + mock fallback, but never runs (EMBEDDINGS_ENABLED=false everywhere).

Not started

- Voice (Web Speech API) — named success metric "behavioural-voice"; zero code.
- Offline ingestion pipeline (clean → chunk → embed → upsert) and n8n workflow — README-only.
- Supabase persistence + Auth + RLS in practice — migrations unapplied; client never imported by any route.
- Final presentation & write-up (Deliverable #3).
- Fit-scorer validation study — **was not started; now BUILT** (see companion PDF).

Mocked / stubbed — not production-real

- pgvector retrieval → in-memory lexical; match_onet_chunks RPC never called.
- Embeddings → deterministic non-semantic hash by default.
- Supabase / persistence / auth → mock client; nothing read or written; fixed mock user id.
- Case/behavioural inputs → always authored fixtures, regardless of mode.
- O*NET corpus → 21 hand-picked occupations / 84 skills, not 1,016.

Risks / blockers

- **Validation study** was the critical-path deliverable and was unstated (now built; result in companion PDF).
- **Voice** has a hard demo dependency and is 0% — lowest effort, must be scheduled now.
- **Real mode is all-or-nothing** — needs Anthropic + all 3 Supabase vars together or it throws.
- **Scope-creep trap** — the verification doc recommends mock+Haiku for the demo; pgvector/Supabase may be unnecessary for July 30.
- **Stale counts** in CLAUDE.md (81) / deployment doc (86) vs actual 103 tests.

Team action checklist (proposal §10 roles)

Member	Actions
Rui (priority)	Validation harness is built — review it; run the full + real-embeddings ablation; build the ingestion pipeline + n8n if real RAG is pursued.
Feroz	Decide demo posture (mock+Haiku vs real Supabase+pgvector); if real, implement the onet-retrieval pgvector path + wire Supabase/Auth; expand the taxonomy.
Emmanuel	Web Speech API voice on behavioural (Chrome + text fallback); file-upload UI (extractText already exists in lib).
Ibuken	More cases/questions; user testing on the live URL; start the write-up; reconcile stale counts.

Critical path — next 7 days

- 1. Validation: **done this pass** — review results, then iterate toward the 70/90 target (see companion PDF).
- 2. Voice spike on behavioural (Emmanuel) — Chrome STT → textarea behind a flag.
- 3. Lock demo posture (Feroz) — recommend mock + Haiku; defer pgvector unless persistence is required.
- 4. Parser robustness number — reuse the validation corpus to report the $\geq 95\%$ parse metric.

Generated from the working audit (you-are-auditing-my-snazzy-sky.md). Figures and numbers reflect the repo on 20 June 2026.